On the other hand, cool air is dense and heavy. As a consequence it sinks vertically. It results in additional weight and pressure which cause a high pressure area to occur on the ground.

6.4.3 Pressure Belts of the Earth

The atmospheric pressure belts envelope on the surface of the earth. They are equatorial low pressure belt, sub tropical high pressure belts, sub polar low pressure belts and polar high pressure belts.

6.4.4 Wind Systems

Wind is the horizontal movement of air molecules from areas of high pressure to areas of low pressure to maintain the atmospheric equilibrium. The wind always moves perpendicular to isobars. If the earth did not rotate, the winds would blow in a straight path. Then the rotation of the earth results in corilois effect and it deflects the direction of the wind. Wind direction is identified by an instrument called Wind Vane and wind speed is measured by Anemometer.

Coriolis Effect

The rotation of the earth affects the moving objects on the earth surface. Free moving objects, affected by the rotation of the earth, do not follow a straight line. In the northern hemisphere they drift towards right and towards left in the southern hemisphere. A car travelling down a straight road at 95 km/hr in northern hemisphere would drift to the right of the path if the friction between surface and tyre is absent. The tendency is called as Coriolis Effect as it was discovered by G.G. Coriolis. This is the reason why racket launching stations are located on the east coastal areas. Example: Sriharikota, French Guyana.

Types of Winds

Winds are classified based on the nature and area of influence as follows;

6.4.5 General Atmospheric Circulation, Pressure Belts and Primary Wind System

From the equator to the poles, each hemisphere has four pressure belts and totally there are seven belts on the globe. The pressure belts lead to formation of primary wind system as follows:

a. **The equatorial low pressure belt (between 5°N and 5°S):** This is the region of calm, weak and changeable winds. Due to the high temperature over this region, the air gets heated expands and become lighter and rises upward and creates low pressure over the region. This region is a belt of calm and referred to as the 'Doldrums'. The winds blow from the sub tropical high pressure belt towards the equatorial low pressure belt. Due to Coriolis Effect these winds are deflected to the right in the northern hemisphere and to the left in the southern hemisphere. As winds are named after the direction from which they originate they are called as the North East and South east trade winds. As the winds favoured trading ships they are called as 'Trade winds'.

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